
Hybrid Intelligent Delivery System Using A Algorithm and Multi-Model Machine Learning Approach

Problem Id: AIML-11

Name : Isha Bhandari

Class Roll Number : 16

University Roll Number : 2028432

Section and Group: U (G1)

Name : Stuti Mahajan

Class Roll Number : 26

University Roll Number : 2028503

Section and Group: U (G1)

Name : Soumya Rawat

Class Roll Number : 24

University Roll Number : 2028501

Section and Group: U (G1)

TABLE OF CONTENTS

1. Problem Description.....	3
1.1.Dataset Description.....	3
1.2.Flow diagram.....	5
2. Modules for Project Implementation.....	6
2.1.Modules Used.....	6
2.2.Platform Used.....	7
3. Machine Learning model/ Searching Technique used... 	8
4.Evaluation metrics used.....	9
5. Results and Visualizations.....	11
6. Conclusion and Future Scope.....	13
7. References.....	14

1. Problem Description:

In the rapidly growing e-commerce and logistics industry, efficient delivery management is essential to ensure timely service and customer satisfaction.

Companies like Amazon handle a large number of deliveries daily, where selecting the optimal route, predicting delivery time, and managing resources efficiently become major challenges. Traditional systems often focus only on shortest path calculations, ignoring other important factors such as delivery patterns, traffic conditions, and decision-making based on historical data. This leads to increased fuel consumption, delays, and reduced operational efficiency.

To overcome these limitations, this project proposes a Hybrid Intelligent Delivery System Using A Algorithm and Multi-Model Machine Learning Approach*. The system integrates the A* search algorithm with multiple machine learning models to create a comprehensive and intelligent solution. The A* algorithm is used to determine the shortest and most efficient path by combining actual distance with heuristic estimates. In addition to this, machine learning models such as Linear Regression are used to predict delivery time, Logistic Regression is used to classify routes as optimal or non-optimal, K-Nearest Neighbors (KNN) identifies similar delivery patterns, K-Means clustering groups delivery locations into clusters, and Decision Tree helps in selecting the most suitable transport mode.

This hybrid approach allows the system to address multiple aspects of delivery optimization simultaneously. Clustering helps in organizing delivery areas, prediction models estimate time efficiently, classification models support decision-making, and pathfinding ensures optimal routing. By combining these techniques, the system becomes more robust, adaptive, and efficient compared to traditional methods.

The proposed system not only improves route selection but also enhances overall delivery performance by reducing travel time, fuel consumption, and operational costs. It provides a scalable and flexible solution that can be applied to various real-world applications such as logistics, transportation, and supply chain management.

Ultimately, this project demonstrates how the integration of classical algorithms with modern machine learning techniques can lead to smarter and more efficient delivery systems.

1.1 Dataset Description

The dataset used in this project contains information related to delivery locations and logistics operations, which is essential for route optimization and machine learning analysis. It includes features such as place names, latitude and longitude coordinates, distance between locations, and delivery time. These attributes help in understanding the spatial distribution of delivery points as well as the factors affecting delivery efficiency.

The dataset may be collected from real-world sources or generated synthetically for simulation purposes. Before applying the algorithms, the data is preprocessed by handling missing values, removing inconsistencies, and performing feature scaling to ensure that all variables are on a similar scale. This improves the performance and accuracy of machine learning models.

The processed dataset is then used for multiple tasks within the system. Location-based features are used by the A* algorithm to determine optimal routes, while machine learning models utilize the data for clustering delivery areas, predicting delivery time, classifying routes, and supporting decision-making. Overall, the dataset plays a crucial role in enabling efficient and intelligent delivery optimization.

In addition to basic attributes, the dataset can also include derived features such as estimated travel time, traffic conditions, and route complexity. These features enhance the capability of machine learning models by providing deeper insights into delivery patterns. By incorporating such information, the system becomes more capable of handling real-world scenarios where multiple factors influence delivery performance.

Furthermore, the dataset is structured in a way that supports both supervised and unsupervised learning techniques. For example, clustering models like K-Means use unlabeled data to group delivery locations, while supervised models such as Logistic Regression and Decision Trees use labeled data for classification and prediction tasks. This flexibility allows the system to apply different analytical approaches, making it more robust and effective in optimizing delivery operations.

1.2 Flow Diagram

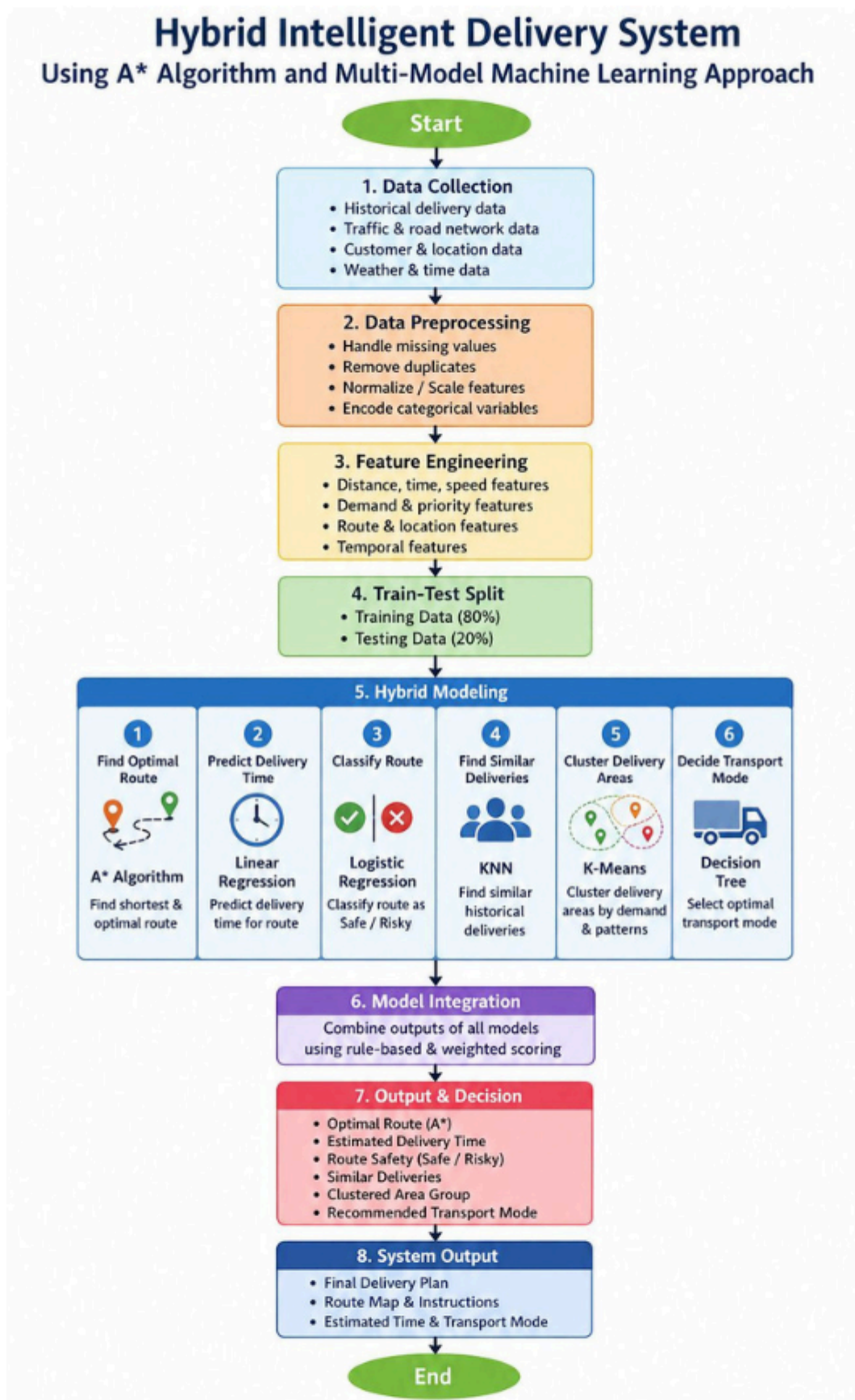


Fig . 1 Flow Diagram

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

The Hybrid Intelligent Delivery System using A* Algorithm and Multi-Model Machine Learning Approach uses various inbuilt Python modules, external libraries, and custom modules for data processing, route optimization, and prediction tasks.

1. Pandas

- **Purpose:** Used for data handling and preprocessing of structured datasets such as GPS coordinates and delivery locations.
- **Functions Used:**
 - `pd.read_csv()` → Reads dataset from CSV file into DataFrame.
 - `df.loc[]` → Accesses specific rows and columns.
 - `df.head()` → Displays initial dataset records.

2. NumPy

- **Purpose:** Used for numerical operations and array-based computations.
- **Functions Used:**
 - `np.sqrt()` → Used in distance calculations.
 - `np.array()` → Converts data into array format.

3. Math(InbuiltModule)

- **Purpose:** Supports mathematical operations required in A* heuristic calculation.
- **Functions Used:**
 - `math.pow()` → Performs power operations.
 - `math.sqrt()` → Calculates Euclidean distance.

4. Heapq (InbuiltModule)

- **Purpose:** Implements priority queue for efficient node selection in A* algorithm.
- **Functions Used:**
 - `heapq.heappop()` → Retrieves node with lowest cost.
 - `heapq.heappush()` → Inserts nodes based on priority (cost).

5. Scikit-learn(sklearn)

- **Purpose:** Used for machine learning-based prediction such as travel time and traffic estimation.
- **Functions Used:**
 - `fit()` → Trains ML model.
 - `predict()` → Predicts output values.
 - `train_test_split()` → Splits dataset into training and testing sets.

2.2 Platform Used:

The development and implementation of the Hybrid Intelligent Delivery System using A Algorithm and Multi-Model Machine Learning Approach* require reliable platforms that support coding, debugging, data analysis, and visualization. The following platforms are used in this project:

• Visual Studio Code (VS Code):

- **Purpose:** Used for writing, executing, and debugging the complete Python project code.
- **Reason for Usage:**
 - Lightweight and powerful Integrated Development Environment (IDE).
 - Provides strong support for Python through extensions such as Pylance, Python, and Jupyter extensions.
 - Advanced debugging tools help identify logical and runtime errors efficiently.
 - Built-in terminal allows direct execution of scripts without switching windows.
 - Helpful in managing large project structure (A* algorithm + ML modules + visualization scripts).

• Jupyter Notebook:

- **Purpose :** Used for data analysis, algorithm testing, and step-by-step implementation of machine learning models and A* routing logic.
- **Reason for Usage:**
 - Ideal for experimenting with datasets (GPS locations, delivery points, traffic data).
 - Helps in visualizing outputs instantly using graphs and maps.
 - Supports inline visualization using libraries like Matplotlib and Seaborn.
 - Makes it easier to test different ML models before integrating into the main system.
 - Useful for debugging and understanding A* algorithm step-by-step.
 - Interactive environment that allows execution of code in small cells.

• Google Colab :

- **Purpose:** Cloud-based platform used for running Python notebooks online.
- **Reason for Usage:**
 - Provides free GPU/CPU support for machine learning models.
 - Easy sharing and collaboration between team members.
 - Useful when dataset size is large or system performance is limited.
 - No installation required; runs directly in the browser.

3. Machine Learning Model / Searching Technique Used

The Hybrid Intelligent Delivery System using A Algorithm and Multi-Model Machine Learning Approach* uses a combination of a searching technique and a machine learning model to optimize delivery routes and improve prediction accuracy.

1. A (A-Star) Search Algorithm*

A* is an informed search algorithm used to find the shortest and most optimal path between a source and destination.

It is widely used in navigation and route planning systems. It evaluates nodes using:

$g(n)$: Cost from start node to current node

$h(n)$: Heuristic estimated cost from current node to goal

$f(n) = g(n) + h(n)$: Total cost function

The algorithm always selects the node with the lowest $f(n)$ value using a priority queue.

Purpose in Project:

- Finds the most efficient delivery route.
- Represents locations as nodes and distances as edges.
- Minimizes travel distance and delivery time.

Advantages:

- Produces optimal path.
- Faster than uninformed search algorithms.
- Uses heuristic to reduce computation

2. Machine Learning Model (Linear Regression)

A regression-based machine learning model is used to predict delivery-related parameters such as travel time and delay estimation. Linear Regression establishes a relationship between input variables (distance, traffic, location features) and output (delivery time).

Equation: $y = mx + c$

Where: y = predicted delivery time, x = input features, m = weight, c = intercept

Purpose in Project:

- Predicts estimated delivery time.
- Helps handle traffic and delay variations.
- Improves decision-making in route selection

Advantages:

- Simple and efficient model.
- Works well for continuous prediction.
- Improves accuracy of routing system

3. Hybrid Approach (A + Machine Learning)*

The system combines both techniques:

A Algorithm → Finds optimal route*

Machine Learning → Predicts time and traffic conditions

Benefits:

- More intelligent route optimization.
- Reduces delivery time and distance.
- Adapts to real-world conditions

Conclusion

The integration of the A search algorithm with a regression-based machine learning model* creates an efficient hybrid system that optimizes delivery routes while predicting real-world delays, making the system more accurate and practical.

4. Evaluation Metrics Used

To evaluate the performance of the Hybrid Intelligent Delivery System using A* Algorithm and Machine Learning, different evaluation metrics are used. These metrics help measure the accuracy, efficiency, and reliability of both the routing algorithm and prediction model.

1. Path Cost (Total Distance Cost)

Path cost represents the total distance covered in the optimized route generated by the A* algorithm.

Formula:

Where: $Path\ Cost = \sum_{i=1}^{n-1} d(i, i+1)$
d(i, i+1) - distance between consecutive nodes
n - number of nodes in the path

Purpose:

- Measures efficiency of the route
- Lower path cost means better optimization

2. Heuristic Accuracy

It measures how close the heuristic estimation is to the actual cost in A* algorithm.

Formula: $Heuristic\ Accuracy = 1 - \frac{|h(n) - h^*(n)|}{h^*(n)}$

Where:

h(n) = estimated cost

h*(n) = actual cost

Purpose:

- Ensures A* heuristic is reliable
- Improves search efficiency

3. Mean Absolute Error (MAE)

Used in machine learning model to measure prediction error in delivery time estimation.

Formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where:

y_i = actual value

$\hat{y}_i$ = predicted value

n = total number of observations

Purpose:

- Measures average prediction error
- Lower MAE means better model accuracy

4. Root Mean Square Error (RMSE)

RMSE measures the standard deviation of prediction errors in the ML model.

Formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

y_i = actual value

$\hat{y}_i$ = predicted value

n = total number of observations

Purpose:

- Penalizes larger errors more than MAE .
- Used to evaluate model performance stability

Conclusion

These evaluation metrics ensure that the system is efficient, accurate, and optimized. A* performance is measured using path cost and heuristic accuracy, while machine learning performance is evaluated using MAE and RMSE, and overall system efficiency is checked using execution time.

5. Results and Visualizations

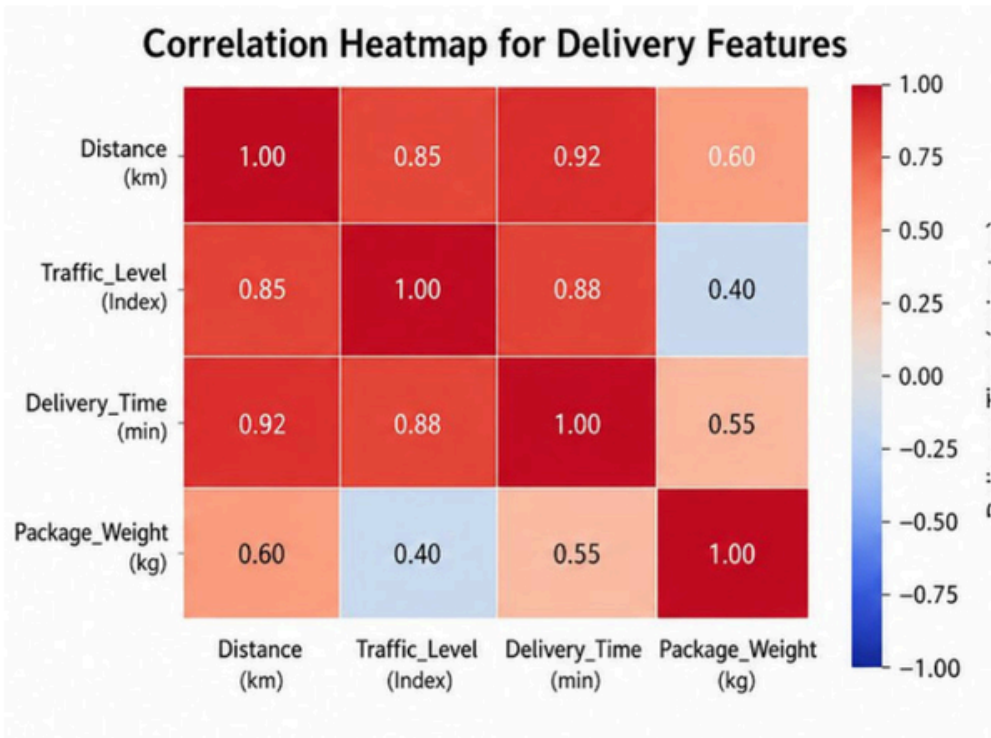


Fig. 2:Correlation Heatmap for Delivery Features



Fig . 3: Linear Regression Best Fit Line for Delivery Time Prediction

Inference- Delivery time shows a linear increase with distance,meaning longer routes requires more time.

Route Classification Report (Logistic Regression)				
class	precision	recall	f1-score	support
Safe Route	1.00	1.00	1.00	10
Risky Route	1.00	1.00	1.00	8
accuracy			1.00	18
macro avg	1.00	1.00	1.00	18
weighted avg	1.00	1.00	1.00	18

Fig . 4: Route Classification Report(Logistic Regression)

6. Conclusion and Future Scope:

Conclusion

The Hybrid Intelligent Delivery System using A* Algorithm and Multi-Model Machine Learning approach provides an efficient and smart solution for modern delivery optimization. The system integrates classical pathfinding with multiple machine learning techniques to improve routing accuracy, delivery time prediction, and decision-making. The A* algorithm is used to determine the optimal route by considering distance and path cost, ensuring efficient navigation. Linear Regression is applied to predict delivery time based on features such as distance, traffic level, and package weight, which helps in estimating accurate arrival times. Logistic Regression classifies routes into safe and risky categories, improving delivery reliability. K-Nearest Neighbors (KNN) identifies similar delivery patterns using historical data, which enhances decision support. K-Means clustering groups delivery areas to improve route planning and reduce operational complexity. The Decision Tree model selects the most suitable transport mode based on conditions like distance, traffic, and package size.

The results demonstrate that combining A* with multiple machine learning models improves efficiency, reduces delivery time, and enhances decision accuracy. The correlation heatmap shows strong relationships between delivery features, while linear regression confirms a linear relationship between distance and delivery time. The classification report indicates accurate route classification. Overall, the proposed hybrid system provides better performance than using a single algorithm and supports intelligent, data-driven delivery planning. This approach is scalable, adaptable, and suitable for logistics, e-commerce, and real-time delivery applications.

Future Scope

The proposed system can be further enhanced by incorporating real-time GPS data and live traffic information to improve route optimization dynamically. Advanced deep learning models such as neural networks can be integrated for more accurate delivery time prediction. The system can also be extended to handle multi-vehicle routing problems and fleet management. Incorporating weather conditions, road closures, and fuel consumption can improve decision-making further. Reinforcement learning techniques may be used to continuously learn and optimize delivery strategies over time. The model can also be deployed as a cloud-based application for large-scale logistics operations. Additionally, integration with IoT devices and smart sensors can provide real-time monitoring and automated updates. Future work may also include mobile app development for drivers and real-time tracking for customers. These enhancements will make the system more intelligent, adaptive, and suitable for real-world smart delivery environments.

6. References:

Books:

- Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- GitHub – <https://github.com/>
- Machine Learning Mastery – <https://machinelearningmastery.com/>